Modality-Driven Search with Holistic Trace Judging for ARC-AGI-2

72.9% verified result, beating frontier models by ~20%

Johan Land

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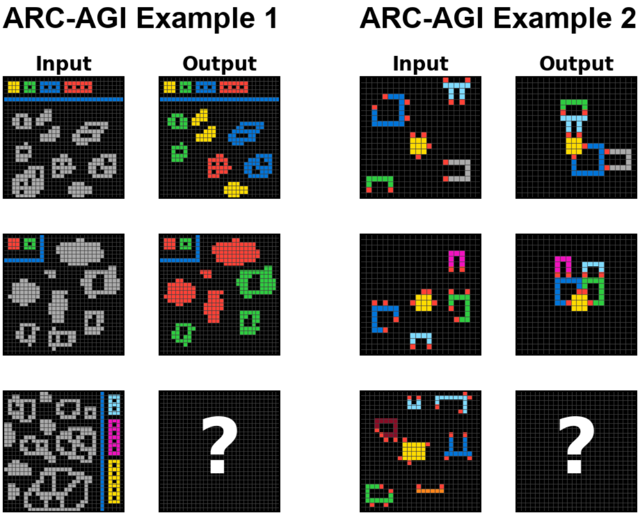
ARC-AGI-2 — a brief history, and two example problems

- **2019** — ARC-AGI-1 introduced (Chollet); near-100% human performance on calibrated tasks, near-zero from contemporary AI systems
- **2020–2023** — Program synthesis era; top scores reach ∼20% on ARC-AGI-1
- **Dec 2024** — o3 saturates ARC-AGI-1 (∼76% at $200/task), but only ∼3% on ARC-AGI-2
- **2025** — Focus shifts to ARC-AGI-2; top scores climb to ∼50% (GPT-5.2, Opus 4.5)
- **Dec 2025** — **This work: 72.9% on ARC-AGI-2**, first verified >70%
- **2026** — Frontier models leapfrog to ∼80%+ (GPT-5.5, Gemini 3.1 Deep Think, Opus 4.7); ARC-AGI-3 launches (<1%)



What is ARC-AGI-2? — format and evaluation

What the model sees (JSON):

```
{
  "train": [
    {"input":  [[0,1,0],[1,0,1],[0,1,0]],
     "output": [[1,0,1],[0,1,0],[1,0,1]]},
    {"input":  [...], "output": [...]},
    ...
  ],
  "test": [
    {"input":  [[1,0,1],[0,1,0],[1,0,1]]}
  ]
}
```

What the model returns:

```
[[0,1,0],[1,0,1],[0,1,0]]
```

- **pass@2:** the model is allowed *two guesses* per test input — it scores if either matches the ground truth exactly
- **Public eval** for development; the **private eval** is the official scoring — run end-to-end by the ARC Prize foundation on tasks the developer never sees
- **Cost per task** (USD) is graded alongside accuracy — efficiency is a first-class metric

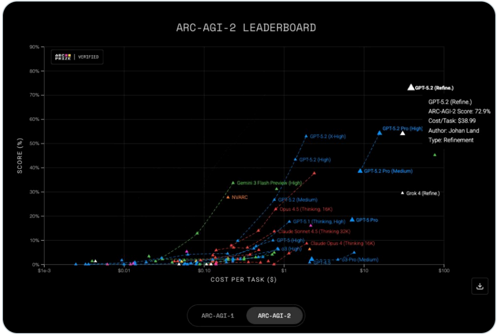
Recognized as new public SOTA by the ARC Prize team

ARC Prize @arcprize · Feb 4

New SOTA public submission to ARC-AGI:

- V1: 94.5%, $11.4/task
- V2: 72.9%, $38.9/task

Based on GPT 5.2, this bespoke refinement submission by @LandJohan ensembles many approaches together

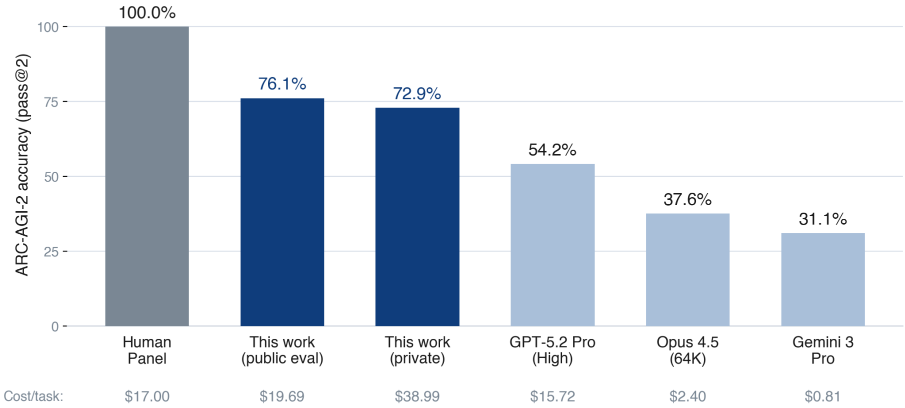


28 · 125 · 912 · 213K

The **ARC Prize Foundation** publicly announced the submission as a **new public SOTA** (Feb 2026).

- *"New SOTA public submission to ARC-AGI"*
- **ARC-AGI-1:** 94.5% at $11.4/task
- **ARC-AGI-2:** 72.9% at $38.99/task
- Independent recognition — announced by the foundation, not self-reported

The headline result



Best publicly reported results as of around Jan 2026, near the time of this submission — the field has moved since.

+18.7 points over the strongest standalone frontier model at the time

Where the leading Chinese models stand

System	Author	Released	ARC-AGI-1	ARC-AGI-2	Cost/Task	Status
This work	—	Dec 2025	**94.5%**	**72.9%**	$38.99	Verified
DeepSeek V4 Pro-Max	*DeepSeek*	Apr 2026	—	*46%*	—	Not verified
Kimi K2.5	Moonshot AI	Jan 2026	65.3%	11.8%	$0.28	Verified
Minimax M2.5	Minimax	Feb 2026	63.7%	4.9%	$0.17	Verified
GLM-5	Z.ai	Feb 2026	44.7%	4.9%	$0.27	Verified
DeepSeek V3.2	DeepSeek	Dec 2025	57.0%	4.0%	$0.12	Verified
Qwen3-235B	Alibaba	Jul 2025	11.0%	1.3%	$0.004	Verified

"Verified" = listed on the official ARC Prize leaderboard. The DeepSeek V4 figure is an unofficial community result.

- Among **officially verified** results the strongest (Kimi K2.5) reaches **11.8%** — most sit **below 5%** (an unofficial DeepSeek V4 figure aside)
- These models are **strikingly cost-efficient** — often 100–1000× cheaper — and competitive on many benchmarks
- But abstract, never-before-seen reasoning is where the gap is widest — and where orchestration pays off

The method is model-agnostic — the same architecture could lift these models too

Two problems: break groupthink, then recognize the outlier

- LLMs produce fluent, internally coherent traces — and are still **confidently wrong**
- On hard tasks, models **cluster around the same wrong interpretation**
- $\Rightarrow$ majority voting / self-consistency *amplifies* the error
- **(1) Break the groupthink** — diversify so some candidate reaches the correct answer
- **(2) Recognize the rare-correct** — pick it out when most others disagree

Roadmap: **(1)** next two slides · **(2)** after that · then both wired together · then on a real task

(1) Break the groupthink — modalities as search operators

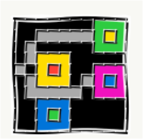
TEXT

```
0,0,0,0,0,0,0,3,3,3
0,0,5,5,5,5,5,3,4,3
0,5,0,0,0,0,0,3,3,3
0,5,4,4,4,0,0,0,0,0
5,0,4,2,4,0,0,6,6,6
0,5,4,4,4,0,5,6,1,6
```

⋮

Reads the grid as numbers / CSV

IMAGE



Sees a rendered PNG of the grid

CODE

```python
def solve(g):
  objs = find_objects(g)
  rule = infer(objs)
  return apply(rule, g)
```

Writes Python to transform the grid

Same task, three representations → three different hypothesis distributions

Each modality solves tasks the others miss · generated **independently** (no shared prompts or scaffolding) · diversity at the *representation* level, not just temperature

(1) Break the groupthink — candidate generation in practice

- **3 models:** GPT-5.2 (x-high), Gemini 3 Preview (high), Claude Opus 4.5 (long-context)
- **3 modality families:** Text, Image, Code
- **29 candidates per task** across families × configurations
- **Deliberately minimal prompts** in every lane *(see slide 10)*
- Code candidates use a sandbox REPL for iterative debugging

(2) Recognize the rare-correct — holistic judging vs. majority voting

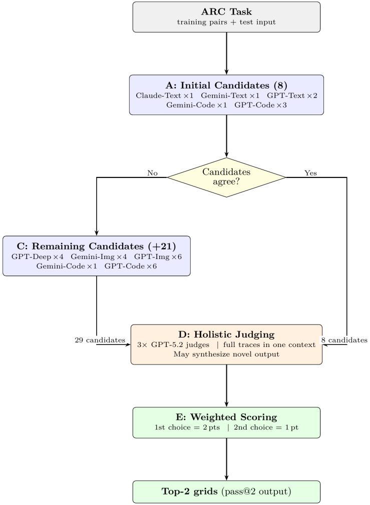
- Concatenate all 29 reasoning traces into one long-context judge prompt
- Each judge picks top-2 candidates *and* explains why other clusters are wrong
- Weighted vote across 3 judges $\longrightarrow$ pass@2 guesses

Net uplift vs. majority-vote baseline: +7 instances

All 7 are minority recoveries — the correct answer wasn't the most common candidate.

Plus **+1 synthesis** instance (`21897d95:2`): zero correct candidates — judge recombined partial insights into a novel correct output.

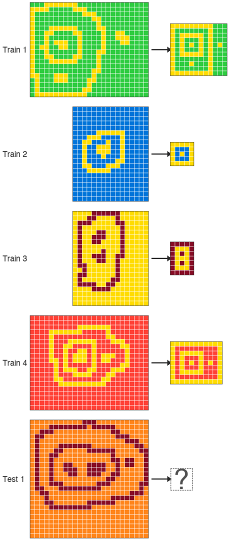
(1) + (2) wired together — pipeline overview



Generate broadly → judge holistically → vote weighted

- Up to 29 candidates across Text / Image / Code, 3 frontier models
- 3 parallel judges read *all* traces in one long-context prompt
- Adaptive early-stop: 37/167 stopped after first probe; **36 correct (97%)**

(1) + (2) on a real task — 28 of 29 candidates wrong, judges picked the right one



Public eval task `2d0172a1:1` — candidate pool $n = 29$

Modality family	Correct	Wrong
Text (8 candidates)	0	8
Image (10 candidates)	**1**	9
Code (11 candidates)	0	11
Total	**1**	**28**

Submission verdict: SOLVED — holistic judges selected the lone correct candidate

- One image candidate produced the correct nested-frame output; 28 of 29 were wrong
- Majority voting would collapse onto the wrong cluster
- **Our judges recovered the minority-correct candidate**

(2) Recognize the rare-correct — what the judges actually read (2d0172a1:1)

This task produced **31,208 lines** of reasoning traces / logs.

Winning candidate

GPT-5.2 x-high, image modality

```
// logs/2026-01-04_22-09-54_2d0172a1_1_step_5.json
{
 "image": {
  ...
  "gpt-5.2-xhigh_13_step_5_image_...": {
   "actual_model": "gpt-5.2-xhigh",
   "input_tokens": 21188,
   "output_tokens": 59323,
   "Full raw LLM response":
"### What the examples show
In every example, the foreground pixels form
several **separate components** of two types:
1. **Frames (loops):** ... every such component
 becomes a **perfect axis-aligned
 rectangular border** (thickness 1).
2. **Markers (solid small components):** ...
 each marker is reduced to a
 **single foreground cell**.
...
### Applying to the test input
- Background color = **7**, foreground = **9**.
- One **outer loop frame**, one **inner loop
 frame** inside it, ... two **markers**
 inside, side-by-side ⇒ **horizontal**.
... outer size **9 rows × 13 columns**.",
   "Extracted grid": [...]
  }, ...
 }, ... }
```

Judge’s selection rationale

GPT-5.2 x-high × 3; all 3 picked correctly

```
// logs/2026-01-04_22-09-54_2d0172a1_1_step_finish.json
{
 ...
 "selection_details": {
  "judges": {
   "duo_pick_council": [
    ..., {
     "model": "gpt-5.2-xhigh",
     "response":
"Across the four solved examples, the
consistent mechanic is **topological
normalization**:
- Each **loop** becomes a **perfect 1-cell-
 thick rectangular frame**.
- Each **solid component** becomes a
 **single foreground cell (a dot)**.
...
The two that best match the examples *and*
the apparent test structure are:
- **Solution 10**: outer frame contains an
 inner frame **plus an extra dot to the
 right** (matching the extra small
 disconnected 9-component at cols
 ~20--21 in the test input) ...
- **Solution 6/7**: the same, but **without**
 the extra right-side dot.
### Candidate 1 (most likely): Solution 10",
     "picked_grids": [...]
    }, ...
   ]
  }
 },
 "result": "PASS" }
```

(1) + (2) across 130 tasks — modality contributions and judges' verdicts

How to read (one row = one of **130 instances** with full 29-candidate coverage):

- **Judges column** (leftmost): the system's final pass@2 verdict — did the holistic judges submit a correct answer?
- Other columns grouped by modality family — **Text** · **Image** · **Code** — each sub-column = one model/config candidate
- **green** = correct · **red** = wrong
- Rows sorted by difficulty: hardest tasks top, easiest bottom

Three bands tell the story:

- **Top:** all-red rows — generation failures (no modality solved them)
- **Middle:** *scattered green cells in mostly-red rows* $\Rightarrow$ different modalities catch different problems
- **Bottom:** mostly green — the easy tasks (many modalities solve)

Edge case — Judges green, all candidates red: a *judge-synthesis* solve. Happens on `21897d95:2`: zero candidates correct, but a judge **recombined partial insights** across failed candidates into a novel correct output.

What the numbers say ($n=130$):

- Exclusive solves: **2 Text-only** · **6 Image-only** · **7 Code-only**

Negative results worth knowing

1. **Prescriptive prompting degrades performance.**\ Templates, CoT scaffolding, domain heuristics — all hurt on the hardest tasks. Mechanism: a *compliance tax on reasoning* (budget spent following the template instead of solving the problem).

2. **Iterative refinement reduces diversity.**\ Refining candidates against each other collapses them onto the same wrong cluster — the exact failure mode we're trying to escape. Same effect from hint-then-solve pipelines.

3. **Pipeline decomposition (objects $\rightarrow$ transformations $\rightarrow$ solver) hurts.**\ Brittle handoffs between stages. Both verbose and terse handovers regress toward the mean rather than expand the hypothesis space.

4. **Strict JSON outputs underperform CSV.**\ Forcing API-level response-format schemas consistently lost to CSV + robust regex parsing. Strict schemas appear to constrain reasoning quality, not just output shape.

5. **Pixel-perfect image renderings hurt.**\ *Slightly distorted* renders beat pixel-perfect ones for image-modality candidates. Clean renders push the model into cell-by-cell numerical reasoning; intentional imprecision forces engagement with shapes, symmetries, and spatial relationships — which is the point of image prompting.

$\Rightarrow$ **Final system: deliberately minimal prompts, independent generation, no decomposition, CSV I/O, deliberately fuzzy image renders.**

Takeaways

(1) Generate broadly — different modalities (text, image, code) and different frontier models, generated *independently*. The hypothesis space matters more than the model.

(2) Judge holistically — a long-context judge reads the *full reasoning*, not just final answers. Majority voting discards the rare-correct; trace-aware judging recovers it (and occasionally synthesizes a new one).

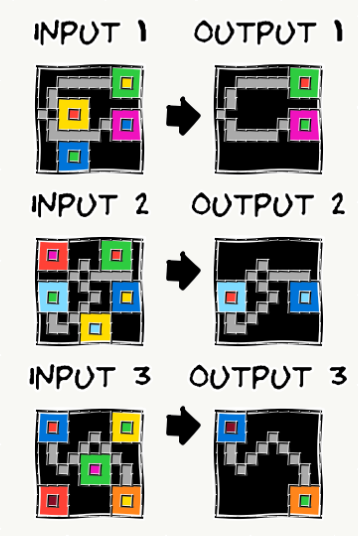
Potential for generalization — other hard problems where the *reasoning*, not just the final answer, signals correctness. Candidate domains:

- Lemma discovery in mathematics
- Algorithm design for novel problems
- Causal inference from observational data
- Reverse engineering
- Scientific hypothesis generation
- Investment-thesis evaluation

Code: `github.com/beetree/ARC-AGI` · Paper QR on title slide

Questions?

End-to-end walkthrough: d35bdbdc



From 29 candidates → 2 guesses

Backup: cost breakdown

- **Public-eval cost is the representative number: $19.69/task**
- Private $38.99/task inflated by GPT-5.2 API instability (84% failure rate, 2,216/14,106 attempts succeeded)
- Tool-integrated code generation is the largest line item
- ZDR mode used in private verification disables tool calls $\longrightarrow$ both lower accuracy and altered cost profile
- ARC-AGI-1 on the same system: 94.5% at $11.40/task — easier tasks early-exit cheaply

Backup: failure decomposition (167 instances)

- **128 solved (76.6%)**
- 21 generation failures (no correct candidate in pool)
- 1 early-stop failure (`dbff022c:1` — extreme groupthink on a legend ambiguity)
- 17 **selection** failures (correct candidate exists, judge missed it)
- 1 synthesis success (`21897d95:2`)

Most future headroom is in selection, not generation.

Backup: why three judges?

- Single judge is biased toward its own prior
- 3 parallel judges + weighted scoring (2 pts first, 1 pt second) hedges *within* the judging step
- Pass@2 format lets the final output also hedge across two interpretations when judges disagree